

Server Security And Maintenance

1. Install the OpenSSH server package on your Linux system and verify that the sshd service is running by connecting to localhost using the ssh command.

ANS :

- ❖ **Install OpenSSH Server**
- ❖ RHEL
 - `sudo yum install openssh-server -y`
- ❖ **Start and Enable SSH Service**
 - `sudo systemctl start sshd`
 - `sudo systemctl enable sshd`
- ❖ **Verify SSH Service Status**
 - `sudo systemctl status sshd`
- ❖ **Test Connection to Localhost**
 - `ssh localhost`
- ❖ Expected Output:
 - The authenticity of host 'localhost' can't be established.
 - Are you sure you want to continue connecting (yes/no)?
- ❖ Type:
 - yes
- ✓ Enter your password to log in.

2. Generate a new SSH key pair using ssh-keygen, and add your public key to the `~/.ssh/authorized_keys` file of your user account to enable key-based authentication.

ANS :

- ❖ Generate a new SSH key:
 - `ssh-keygen -t rsa -b 4096`
- ❖ Press Enter to accept the default location:
 - `/home/user/.ssh/id_rsa`
- ❖ Files created:
 - `~/.ssh/id_rsa`
- ❖ View public key:
 - `cat ~/.ssh/id_rsa.pub`
- ❖ Add public key to authorized_keys:
 - `cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys`
- ❖ Set correct permissions:
 - `chmod 700 ~/.ssh`

- ❖ Test key authentication:
 - ssh localhost
 - ✓ The login should occur without requiring a password.
3. Edit the /etc/ssh/sshd_config file to disable password authentication by setting PasswordAuthentication no, then restart the sshd service and test that you can only log in using your SSH key.

ANS :

- ❖ Open SSH configuration file:
 - sudo nano /etc/ssh/sshd_config
 - ❖ Locate:
 - #PasswordAuthentication yes
 - ❖ Change to:
 - PasswordAuthentication no
 - PubkeyAuthentication yes
 - ✓ Save and exit.
 - ✓ Restart SSH service:
 - sudo systemctl restart sshd
 - ❖ Verify configuration:
 - sudo systemctl status sshd
 - ❖ Test SSH login:
 - ssh localhost
 - ✓ Successful login
4. Document each step you took to configure SSH key-based authentication and disable password login, including the exact commands and configuration changes you made.

Hint: Use screenshots or command outputs to support your documentation.

ANS :

- ❖ Commands Used:
 - sudo yum install openssh-server -y
 - sudo systemctl start sshd
 - sudo systemctl enable sshd
 - sudo systemctl status sshd
 - ssh localhost
 - ssh-keygen -t rsa -b 4096
 - cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
 - chmod 700 ~/.ssh

- `chmod 600 ~/.ssh/authorized_keys`
- `sudo nano /etc/ssh/sshd_config`
- `sudo systemctl restart sshd`
- `ssh localhost`

❖ Configuration Changes:

- `/etc/ssh/sshd_config`
- Modified settings:
- `PasswordAuthentication no`
- `PubkeyAuthentication yes`

❖ Result:

- SSH service installed and running.
- SSH key pair generated successfully.
- Public key added to `authorized_keys`.
- Password authentication disabled.
- Login possible only through SSH key authentication.